

MODULE 3

Network Layer

Network Layer (OSI Layer 3)

The **Network Layer** is responsible for **end-to-end delivery of data packets** across multiple interconnected networks. Unlike the Data Link Layer, it allows communication between devices on **different networks**.

Functions of the Network Layer

1. Logical Addressing

- Assigns unique identifiers such as **IP addresses**.
- Identifies both the **host** and the **network**.

2. Routing

- Determines the **optimal path** from source to destination.
- Uses algorithms like **Dijkstra's Shortest Path** and **Bellman-Ford**.

3. Packet Forwarding

- Transfers packets from an incoming router interface to the correct outgoing interface.

4. Fragmentation and Reassembly

- Breaks large packets into smaller fragments based on **MTU size**.
- Reassembles fragments at the destination.

5. Error Handling and Congestion Control

- Detects errors and informs upper layers.
- Helps manage congestion and improve network performance.

Packet Switching

Packet Switching is a technique where data is divided into **small packets** and sent independently through the network.

How Packet Switching Works

1. Segmentation of Data

- Large data (email, video, web page) is broken into packets.
- Packet size usually ranges from **512 to 1500 bytes**.

2. Packet Structure

- **Header** contains:
 - Source IP address
 - Destination IP address
 - Sequence number (for ordering)
 - Protocol type (TCP/UDP)
- **Payload**: Actual data

3. Independent Routing

- Each packet travels independently.
- Routers decide paths dynamically based on network conditions.

4. Reassembly at Destination

- Packets are reordered using sequence numbers.
- Missing packets can be detected and retransmitted (in connection-oriented protocols).

Example

- A file is split into 500 packets.
- Packets may travel through different routes (Europe, Pacific, etc.).
- Destination reassembles packets to recreate the original file.

Advantages of Packet Switching

- Efficient resource utilization
- High resilience (rerouting during failures)
- Scalable for large networks
- Cost-effective (no dedicated path required)

Disadvantages of Packet Switching

- Variable packet delay (latency)
- Possible packet loss

- Out-of-order packet delivery

Virtual Circuit Networks (Connection-Oriented)

A **virtual circuit** establishes a **fixed logical path** before data transmission begins.

Phases of Virtual Circuit

1. Connection Setup Phase

- Logical connection established between source and destination.
- Routing tables configured along the path.

2. Data Transfer Phase

- Packets carry a **Virtual Circuit Identifier (VCI)**.
- All packets follow the same predefined path.

3. Connection Teardown Phase

- Connection is terminated after data transfer completes.

Example

- Bank ATM network
- Secure data (PIN verification, transactions) follows the same path

Advantages of Virtual Circuits

- Guaranteed in-order packet delivery
- Predictable delay and performance
- Easier flow control and error handling

Disadvantages of Virtual Circuits

- Initial setup overhead
- Inflexible during congestion or topology changes

Datagram Networks (Connectionless)

Datagram networks send packets **without establishing a prior connection**.

How Datagram Networks Work

- Each packet is treated independently.
- Full source and destination IP addresses included in every packet.
- No setup or teardown required.
- Packets may follow different paths.

Example

- Browsing a website
- HTTP requests and responses sent as independent IP packets
- TCP reorders packets at the destination

Advantages of Datagram Networks

- No connection setup overhead
- Highly flexible and scalable
- Ideal for bursty and short communications

Disadvantages of Datagram Networks

- Packets may arrive out of order
- Packet loss possible (no built-in retransmission at network layer)
- Unpredictable delay

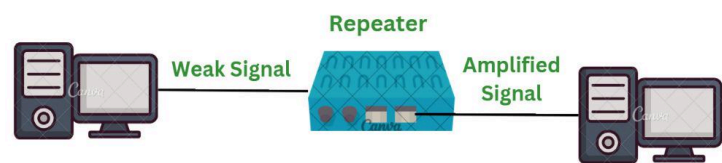
Virtual Circuits vs Datagram Networks

Feature	Virtual Circuit	Datagram Network
Connection	Required	Not required
Path	Fixed	Dynamic
Packet Order	Guaranteed	Not guaranteed
Delay	Predictable	Variable
Flexibility	Low	High

Networking and Internetworking devices



Repeaters



A Repeater is a network device that operates at the Physical Layer (Layer 1) of the OSI model. Its primary function is to regenerate and retransmit signals that have become weak or distorted due to long-distance transmission.

Key Features

- **Signal Regeneration:** Restores weak signals to their original strength without altering the data.
- **Extends Network Range:** Allows data to travel over longer distances by boosting signal strength.
- **Bit-by-Bit Transmission:** Copies and forwards data bits without interpreting them.
- **Transparent Operation:** Does not filter or route data; simply repeats it.

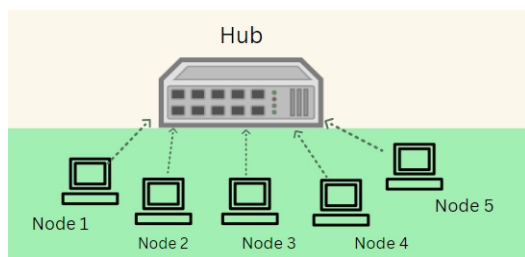
Types of Repeaters

Type	Description
Analog Repeater	Amplifies analog signals; used in older networks and trunk lines.
Digital Repeater	Regenerates digital signals (0s and 1s); used in modern LAN/WAN setups.
Wired Repeater	Used in wired LANs to extend cable length.
Wireless Repeater	Used in Wi-Fi and cellular networks to extend wireless coverage.
Local Repeater	Connects LAN segments separated by short distances.
Remote Repeater	Connects LAN segments across long distances.

Attributes

Attribute	Repeater
OSI Layer	Layer 1 (Physical Layer)
Function	Regenerates and retransmits signal
Filtering	✗ No filtering
Routing Capability	✗ None
Use Case	Extending LAN/WAN range

Hub



A hub is a basic networking device that connects multiple computers or network devices in a Local Area Network (LAN). It operates at the Physical Layer (Layer 1) of the OSI model.

Key Features

- **Multiport Repeater:** A hub is essentially a multiport repeater that receives data and broadcasts it to all connected devices.
- **Broadcasts Data:** It does not filter or route data; every incoming packet is sent to all ports.
- **No Intelligence:** Hubs do not have the capability to determine the destination of data packets.
- **Single Collision Domain:** All devices connected to a hub share the same collision domain, which can lead to network inefficiencies.

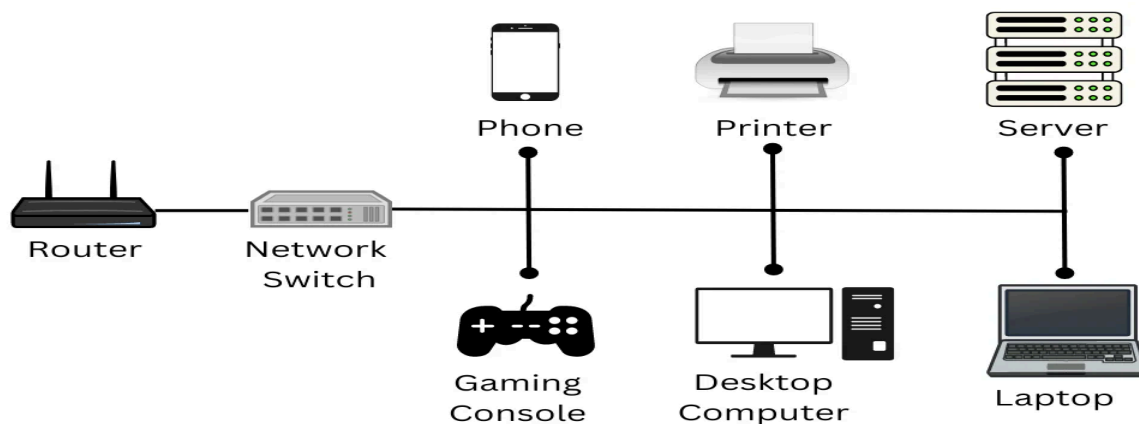
Types of Hubs

Type	Description
Active Hub	Regenerates and strengthens signals before forwarding.
Passive Hub	Simply connects devices without signal regeneration.
Intelligent Hub	Includes management features like monitoring and configuration.

Attributes

Attribute	Hub
OSI Layer	Layer 1 (Physical Layer)
Function	Broadcasts data to all ports
Filtering	✗ No filtering
Routing Capability	✗ None
Collision Domain	Single shared domain
Use Case	Small LANs, legacy networks

Switch



A Switch is a network device that operates at the Data Link Layer (Layer 2) of the OSI model. It connects multiple devices within a LAN and uses MAC addresses to forward data only to the intended recipient device.

Key Features

- **MAC-Based Forwarding:** Uses MAC addresses to direct data to the correct device.
- **Reduces Traffic:** Sends data only to the destination port, unlike hubs.
- **Creates Collision Domains:** Each port is its own collision domain, improving efficiency.
- **Supports Full Duplex:** Allows simultaneous send/receive operations.

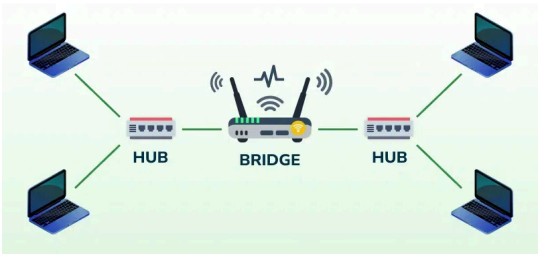
Types of Switches

Type	Description
Unmanaged Switch	Plug-and-play; no configuration needed; used in small networks.
Managed Switch	Configurable; supports VLANs, QoS, SNMP; used in enterprise networks.
Layer 2 Switch	Operates at Data Link Layer; forwards based on MAC addresses.
Layer 3 Switch	Adds routing capabilities; operates at Network Layer.
PoE Switch	Provides power to devices like IP cameras and phones via Ethernet.

Attributes

Feature	Switch
OSI Layer	Layer 2 (or 3)
Data Filtering	✔ Yes (MAC-based)
Collision Domains	One per port
Intelligence	Medium to High
Use Case	LAN segmentation

Bridges



A Bridge is a device used to connect multiple LAN segments into a larger LAN. It operates at the Data Link Layer (Layer 2) of the OSI model and uses MAC addresses to filter and forward data between segments.

Key Functions

- Traffic Filtering: Examines incoming frames and forwards them only if necessary.
- Collision Domain Separation: Each segment connected by a bridge has its own collision domain.
- Bandwidth Optimization: Reduces unnecessary traffic and improves performance.

- **MAC Learning:** Builds a table of MAC addresses to determine where to forward frames.

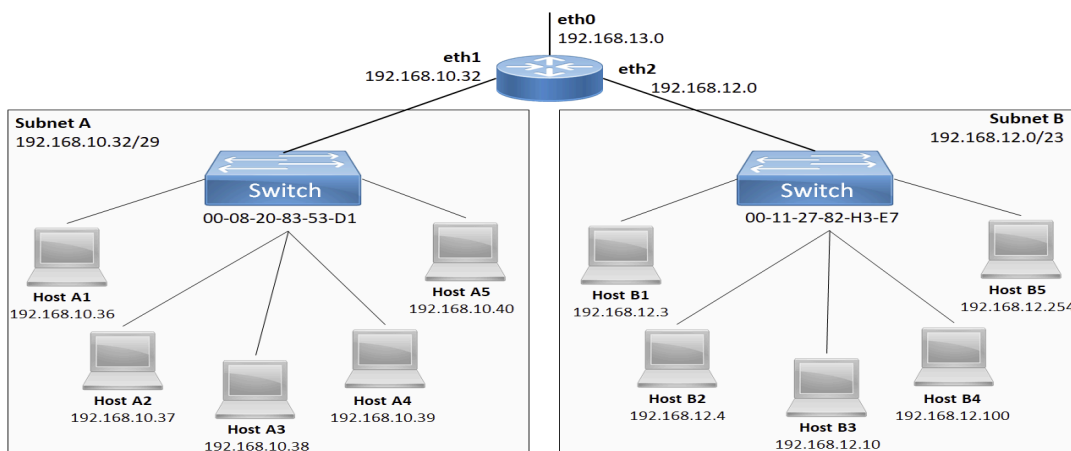
Types of Bridges

Type	Description
Transparent Bridge	Invisible to devices; forwards or blocks frames based on MAC address.
Source Routing Bridge	Developed by IBM for Token Ring networks; uses embedded routing info.
Translational Bridge	Converts data between different network types (e.g., Token Ring ↔ Ethernet).
MAC-Layer Bridge	Filters and repeats packets for similar network segments.

Attributes

Attribute	Bridge
OSI Layer	Layer 2 (Data Link Layer)
Function	Filters and forwards frames
Filtering	✅ Yes (MAC-based)
Routing Capability	❌ No
Use Case	Connecting LAN segments

Router



A Router is a device that operates at the Network Layer (Layer 3) of the OSI model. It is responsible for routing data packets between different networks that use the same communication protocols based on their IP addresses.

Key Features

- **Packet Forwarding:** Determines the best path for data packets using routing tables.
- **Inter-network Communication:** Connects different LANs, WANs, or the internet.
- **Broadcast Domain Separation:** Each interface on a router creates a separate broadcast domain.
- **Traffic Filtering:** Can apply rules to control traffic flow and enhance security.
- **Dynamic Routing:** Supports protocols like RIP, OSPF, and BGP to adapt to network changes.

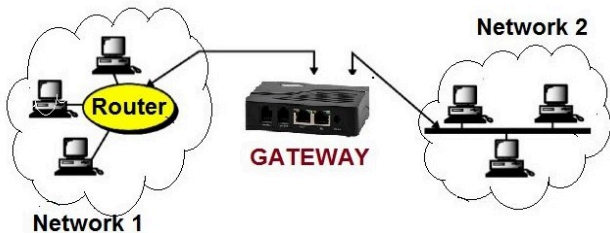
Types of Routers

Type	Description
Wired Router	Connects devices via Ethernet cables; common in homes and offices.
Wireless Router	Combines routing with Wi-Fi access point; used in home networks.
Core Router	High-capacity router used in the backbone of large networks.
Edge Router	Connects internal networks to external networks (e.g., ISP).
Virtual Router	Software-based router used in cloud or virtual environments.

Attributes

Attribute	Router
OSI Layer	Layer 3 (Network Layer)
Function	Routes packets between networks
Filtering	✔ Yes (IP-based)
Routing Capability	✔ Yes
Use Case	Connecting LAN to WAN/Internet

Gateway



A Gateway is a device that connects two different networks that use different communication protocols. It acts as a translator, enabling data exchange between systems that otherwise couldn't communicate. Gateways can operate at any layer of the OSI model, though they typically function at the Network Layer (Layer 3) or higher.

Key Features

- Protocol Translation: Converts data formats and communication protocols between networks.
- Network Interfacing: Connects enterprise networks to external networks like the internet.
- Security Enforcement: Can act as a firewall or proxy server to filter and secure traffic.
- Application Access: Enables access to specialized services like VoIP, cloud storage, or IoT platforms.
- Traffic Direction: Routes data between incompatible systems or applications.

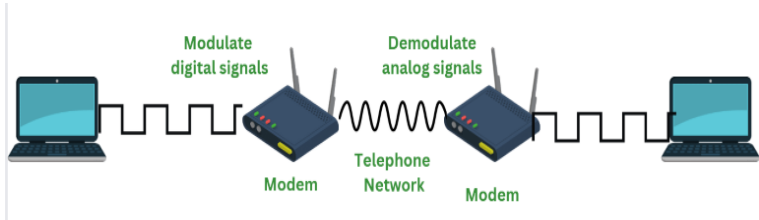
Types of Gateways

Type	Description
Network Gateway	Connects LAN/WAN to the internet; translates IP protocols.
VoIP Gateway	Converts analog voice signals to digital VoIP format.
Cloud Storage Gateway	Translates cloud APIs to file/block storage formats.
IoT Gateway	Bridges IoT protocols (e.g., Zigbee, Bluetooth) to IP networks.
Application Gateway	Manages access to specific applications or services.
Cellular Gateway	Provides internet access via 4G/5G cellular networks.

Attributes

Attribute	Gateway
OSI Layer	Layer 3 and above
Function	Protocol translation & routing
Filtering	✔ Yes (customizable)
Routing Capability	✔ Yes
Use Case	Connecting dissimilar networks

Modem



A Modem (short for Modulator-Demodulator) is a device that converts digital signals from a computer into analog signals for transmission over telephone or cable lines—and vice versa. It enables communication between digital devices and analog transmission media, making it essential for internet access.

Key Features

- **Modulation:** Converts digital data into analog signals for transmission.
- **Demodulation:** Converts incoming analog signals back into digital data.
- **Signal Translation:** Acts as a translator between the ISP and the user's network.
- **Internet Access:** Connects a home or office network to the internet via an ISP.

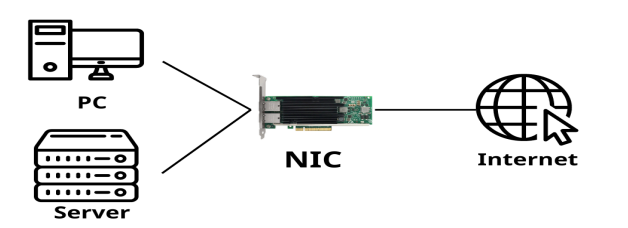
Types of Modems

Type	Description
DSL Modem	Uses telephone lines for internet; supports high-speed broadband.
Cable Modem	Uses coaxial cable TV lines for internet access.
Fiber Modem	Uses optical fiber for ultra-fast internet; often integrated with routers.
Dial-Up Modem	Legacy modem using analog phone lines; very slow.
Wireless Modem	Connects to cellular networks (3G/4G/5G); used in mobile setups.
Satellite Modem	Communicates via satellite; used in remote areas.

Attributes

Attribute	Modem
OSI Layer	Layer 1 (Physical Layer)
Function	Converts digital ↔ analog signals
Filtering	✗ No filtering
Routing Capability	✗ None (unless combined with router)
Use Case	Connecting to ISP/internet

Network Interface Card (NIC)



A Network Interface Card (NIC) is a hardware component that allows a computer or device to connect to a network. It provides the physical interface for communication over wired or wireless networks. NICs are also known as:

- Network adapter
- Ethernet card
- LAN adapter
- Network interface controller

Every NIC has a unique MAC address used to identify the device on a network.

Key Features

- Data Transmission: Sends and receives data packets over the network.
- MAC Addressing: Uses a unique Media Access Control (MAC) address for identification.
- Signal Conversion: Converts data between parallel and serial formats.
- Protocol Handling: Supports Ethernet, Wi-Fi, and other network protocols.
- Status Monitoring: LED indicators show connection and activity status

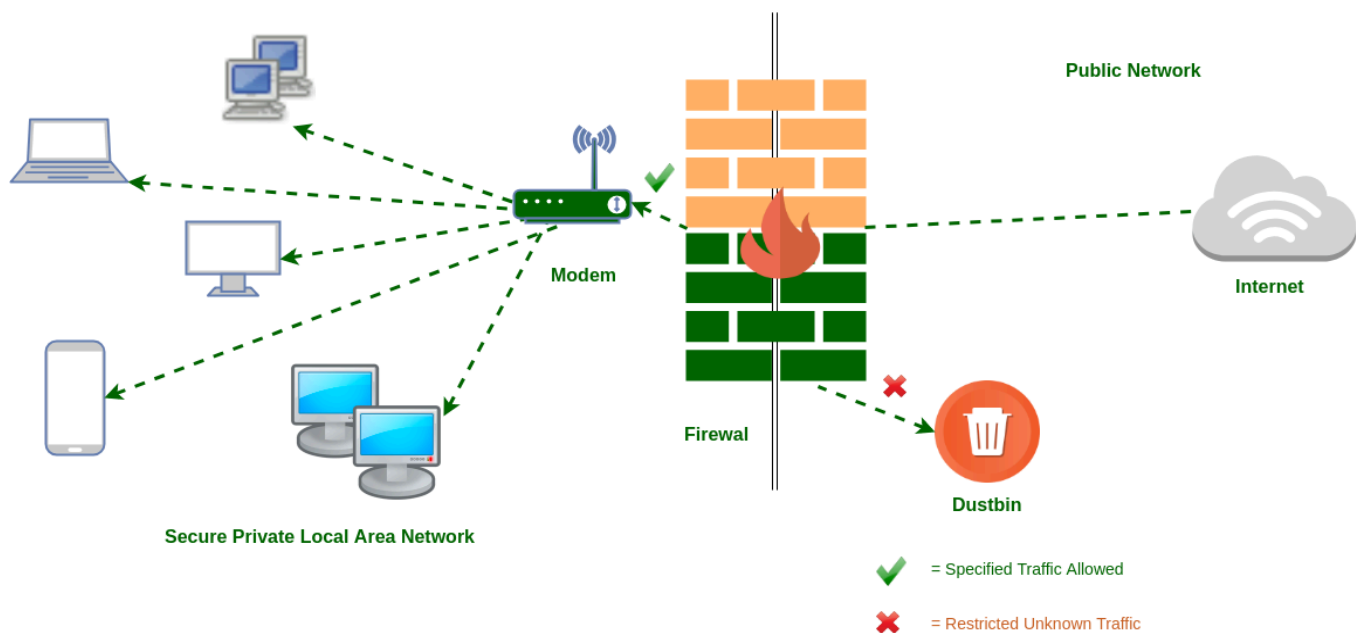
Types of NICs

Type	Description
Wired NIC	Uses Ethernet cables; stable and secure connection.
Wireless NIC	Connects via Wi-Fi; ideal for mobile setups.
Internal NIC	Installed on the motherboard (PCI, PCIe, or onboard).
External NIC	USB or PCMCIA-based; used when internal NIC is unavailable.
Fiber NIC	Uses fiber optic cables for high-speed data transfer.

Attributes

Attribute	NIC
OSI Layer	Layer 1 & Layer 2
Function	Network access and data transmission
Filtering	✗ No filtering
Routing Capability	✗ None
Use Case	Connecting devices to LAN/Wi-Fi

Firewall



A firewall is a network security device—either hardware, software, or both—that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet.

Key Features

- Traffic Filtering: Allows or blocks data packets based on rules.
- Access Control: Prevents unauthorized access to internal systems.
- Threat Detection: Identifies and blocks malware, phishing, and DoS attacks.
- Logging & Monitoring: Keeps records of traffic and alerts on suspicious activity.
- Policy Enforcement: Applies organizational security policies across the network.

Types of Firewalls

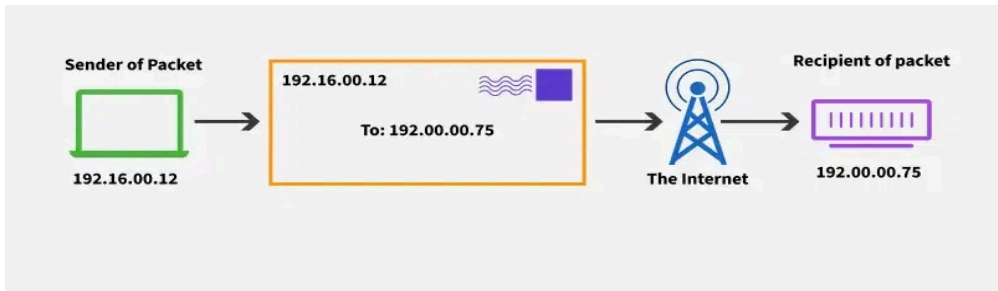
Type	Description
Packet-Filtering Firewall	Filters traffic based on IP, port, and protocol. Basic but fast.
Stateful Inspection Firewall	Tracks active connections and makes decisions based on state and context.
Proxy Firewall	Intercepts and inspects traffic at the application layer.
Next-Gen Firewall (NGFW)	Combines traditional firewall with advanced features like intrusion prevention.
Software Firewall	Installed on individual devices; protects host systems.
Hardware Firewall	Physical device placed at network perimeter; ideal for enterprise setups.
Cloud Firewall (FWaaS)	Firewall as a Service; scalable and centrally managed.

Attributes

Attribute	Firewall
OSI Layer	Layer 3 (Network) & Layer 4 (Transport)
Function	Traffic filtering and access control
Filtering	✔ Yes
Routing Capability	✘ No
Use Case	Network perimeter defense

Internet Protocol (IP)

The Internet Protocol (IP) is a set of rules that controls how data is sent and received over the internet. It assigns a unique IP address to each device and ensures data reaches the correct destination. IP plays a key role in enabling different devices to communicate over the internet, even if they are located in different parts of the world.



Features of Internet Protocol

IP helps in sending data across networks by following these steps:

- **Connectionless Protocol:** IP does not establish a dedicated path between sender & receiver
- **Dividing Data into Packets:** Large data is broken down into smaller packets. This makes it easier and faster to transmit over the network.
- **Addressing:** Each packet carries the sender's and receiver's IP addresses, helping it reach the right destination. This is known as Logical Addressing.
- **Routing the Packets:** Routers read the IP addresses and decide the best path to forward each packet across the network.
- **Reassembling the Data:** At the destination, the packets are reassembled in the correct order to rebuild the original data.
- **Handling Missing Packets:** If some packets are lost or corrupted, other protocols (like TCP) request the missing packets again to complete the data.

Logical Addressing

Logical addressing is a way for computer networks to identify devices using an IP address. This address is unique and can change if the device connects to a different network. Routers use it to send data to the right place. It's different from a MAC address, which is built into the device and never changes.

What is an IP address?

IP address stands for Internet Protocol (IP) address. It is used to identify the network device connected to the internet. The Internet sends information using an IP address. IP address is assigned by the Internet Service Provider (ISP) or Network Administrator, which varies based on time and location.

It serves two main purposes:

- **Identification (Network ID):** It uniquely identifies a device on a network.
- **Location Addressing (Host ID):** It indicates where a device is located within a network, making data routing possible.

Logical Vs Physical Address (IP Vs MAC)

Feature	MAC Address	IP Address
Layer of Operation	Data Link Layer(Layer 2)	Network Layer(Layer 3)
Scope of Use	Local Network(LAN)	Communication across networks
Address Format	48-bit(6-byte) Hexadecimal	Series of four decimal number
Assignment Process	Assigned by manufacturer	Assigned dynamically or Manually
Functionality	Local delivery of frames within the same network segment	Logical addressing and routing between different networks
Network Layer VS. Data Link Layer	Data Link Layer (Layer 2)	Network Layer(Layer 3)
Examples	00:1A:2B:3C:4D:5E	192.168.1.1

House Analogy: Mailing a Letter

Imagine you're sending a letter to a friend who lives in an apartment building.

Logical Address = Apartment Number + Street Address

This is like the IP address. It tells the postal service where to deliver the letter — the building and the specific apartment.

- Example: Flat 5B, 123 Main Street
- Can be changed if your friend moves to a new apartment
- Used by the postal system (network) to route the letter

Physical Address = Person's Name

This is like the MAC address. It identifies who the letter is for inside the apartment.

- Example: John Doe
- Unique to the person (like a MAC address is unique to a device)
- Doesn't change even if they move to a new apartment

Types of IP Addresses

I. Based on Function

1. Private IP Address

- Used **within** a local/private network (LAN).
- **Not routable** on the internet.
- Assigned manually or by a **DHCP** server.
- Used by home routers, office networks, etc.

2. Public IP Address

- Assigned by **ISPs** to devices accessible over the internet.
- **Unique** globally.
- Used to identify a device or server on the internet.

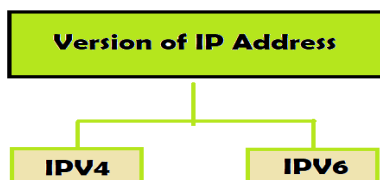
3. Static IP Address

- Manually assigned.
- Does not change unless reconfigured.
- Often used by servers, routers, and networked printers.
- Reliable for hosting websites or remote access.

4. Dynamic IP Address

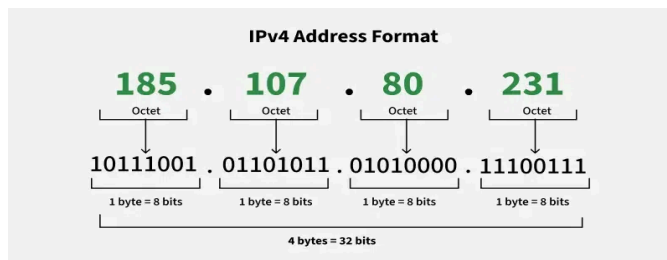
- Automatically assigned by DHCP.
- Can change over time.
- Used for regular devices like laptops, phones.
- Easier to manage for large networks.

II. Based on Version



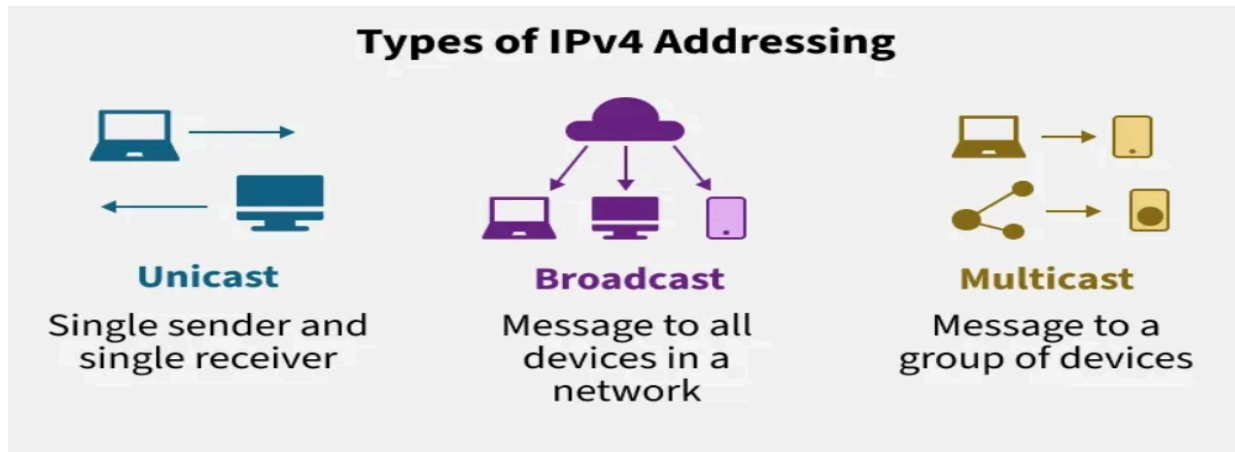
IPV4 (Internet Protocol Version 4)

IPv4 stands for Internet Protocol Version 4. The IPv4 is one of the most commonly used unique addresses for device identification. The IPv4 consists of 32 bits, and it is denoted in a sequence of four numbers separated by "." (dot/period). The period divides the IPv4 address into four octets (an octet is of 8 bits, so 4 octets make up 32 bits). An octet's value ranges from 0 to 255. So, an IPv4 can produce around 4.3 billion possible unique addresses.



Types of IPv4 Addressing

IPv4 basically supports three different types of addressing modes:

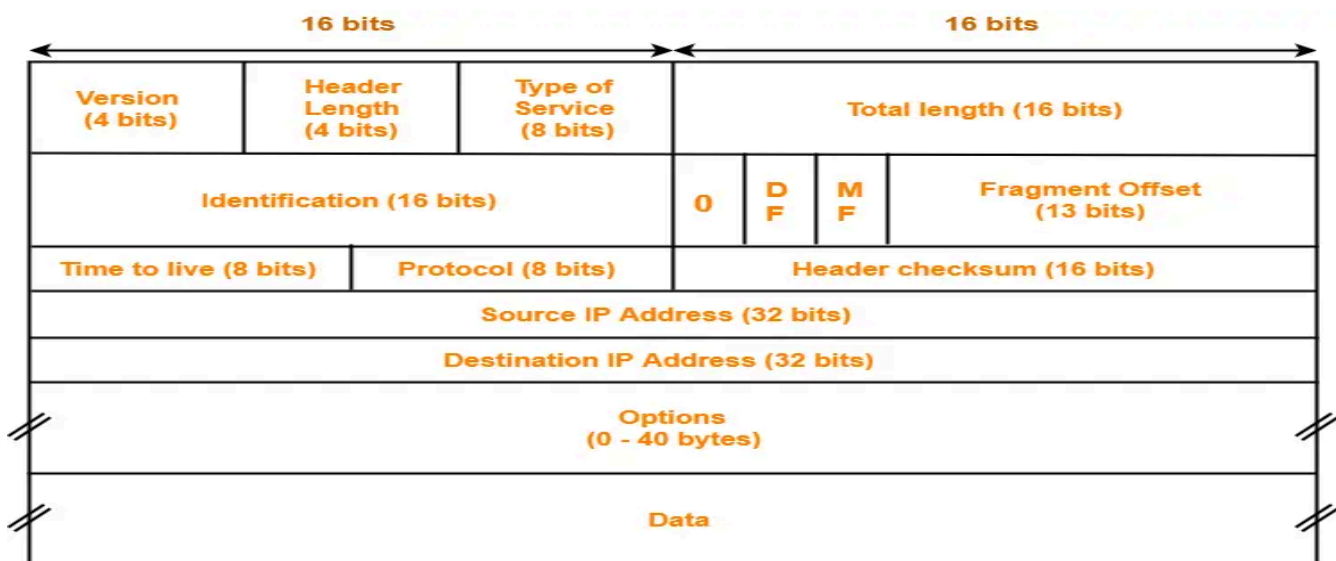


- **Unicast Addressing Mode:** This addressing mode is used to specify single sender and single receiver. Example: Accessing a website.
- **Broadcast Addressing Mode:** This addressing mode is used to send messages to all devices in a network. Example: sending a message in local network to all the devices.
- **Multicast Addressing Mode:** This addressing mode is typically used within a local network or across networks and sends messages to a group of devices. Example: Streaming audio to multiple devices at once.

Datagram

Packets in the IPv4 layer are called datagrams. A datagram is a variable-length packet consisting of two parts: header and data. The header is 20 to 60 bytes in length and contains information essential to routing and delivery.

IPv4 Header Format



IPv4 Header

Version (VER)

- **Size:** 4 bits
- **Value:** Always 4 for IPv4
- **Purpose:** Identifies the IP version (IPv4 or IPv6)

Header Length (HLEN)

- **Size:** 4 bits
- **Unit:** Measured in 4-byte blocks
- **Range:** 5 to 15 (→ 20 to 60 bytes)
- **Purpose:** Indicates where the actual data starts
- **Note:** Minimum header = 20 bytes; max = 60 bytes (with options)

Type of Service (ToS)

- **Size:** 8 bits
- **Purpose:** Sets Quality of Service (QoS) preferences
- **Examples:** Low Delay, High Throughput, High Reliability

Total Length

- **Size:** 16 bits
- **Formula:** Header Length + Payload Length
- **Range:** 20 to 65,535 bytes
- **Purpose:** Total size of the IP packet

Identification

- **Size:** 16 bits
- **Purpose:** Assigns a unique ID to each IP packet so that **fragments** of the same packet can be correctly reassembled

Flags

- **Size:** 3 bits
- First bit : Reserved bit, always 0 (not used)
- Remaining 2 bits:

Flag	Size	Meaning
DF (Do Not Fragment)	1 bit	1 = Don't fragment; 0 = Fragment allowed
MF (More Fragments)	1 bit	1 = More fragments follow; 0 = Last or only fragment

Fragment Offset

- **Size:** 13 bits
- **Purpose:** Shows the fragment's position in the original packet

Time To Live (TTL)

- **Size:** 8 bits
- **Purpose:** Limits packet lifetime by counting how many **hops** (routers) it can pass through

Protocol

- **Size:** 8 bits
- **Purpose:** Identifies the next-level protocol
- **Examples:**
 - TCP → 6
 - UDP → 17

Header Checksum

- **Size:** 16 bits
- **Purpose:** Detects errors in the header
- **Note:** Recalculated at each router hop; excludes data

Source IP Address

- **Size:** 32 bits
- **Purpose:** IP address of the sender

Destination IP Address

- **Size:** 32 bits
- **Purpose:** IP address of the receiver

Options (Optional)

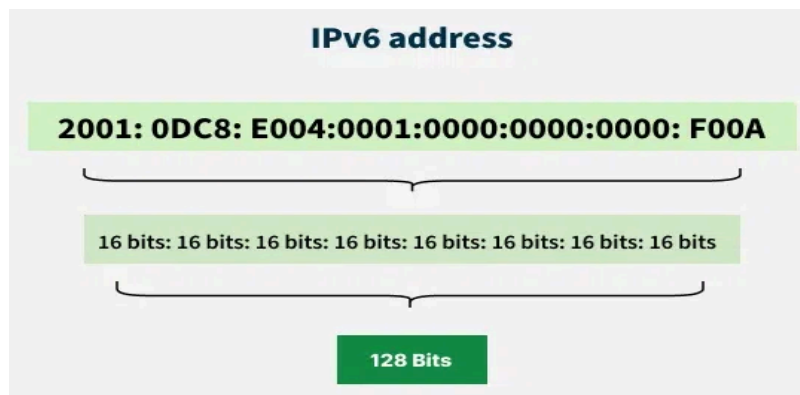
- **Size:** 0 to 40 bytes
- **Purpose:** Adds extra features
- **Examples:**
 - **Record Route:** Logs router IPs along the path
 - **Source Routing:** Specifies the exact route
 - **Padding:** Ensures header is a multiple of 32 bits

Advantages	Disadvantages
Simple and efficient addressing	Limited address space
Wide adoption and compatibility	Complex network configuration
Quality of Service (QoS) support	Inefficient routing and addressing
Network Address Translation (NAT) support	Limited security features

IPv6 (Internet Protocol Version 6)

IPv6 (Internet Protocol version 6) is the most recent version of the Internet Protocol (IP), which is used to identify and locate devices on a network and route traffic across the Internet.

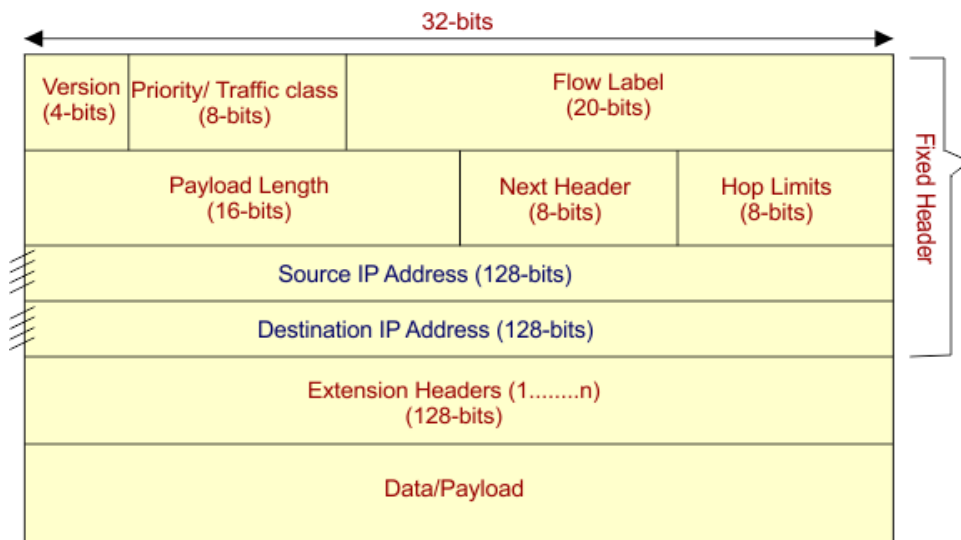
IPv6 has a 128-bit address and has a larger address space (2^{128}) available for future allocation. The 128-bit address is broken into 8 groups, each containing 16 bits. Four hexadecimal numbers represent each group, and colons are used to divide each group from the others.



Types of IPv6 Addressing

Type	Address Range	Delivery Target	Use Case Example
Unicast	2000::/3 , fe80::/10 , fc00::/7	One specific device	Web browsing, file transfer
Multicast	FF00::/8	Multiple group members	Streaming, routing updates
Anycast	Same as Unicast	Nearest device	DNS, CDN, service discovery

IPv6 Header Format



IPv6 Header

The **IPv6 header** is always **40 bytes (320 bits)** long and contains the following fields:

Field	Size	Description
Version	4 bits	Indicates the IP version (always 6 for IPv6).
Traffic Class	8 bits	Specifies packet priority and congestion handling (similar to QoS).
Flow Label	20 bits	Identifies packets belonging to the same flow (e.g., video stream).
Payload Length	16 bits	Size of the payload (excluding the header).
Next Header	8 bits	Indicates the type of the next header (e.g., TCP, UDP, or extension header).
Hop Limit	8 bits	Similar to TTL in IPv4; decremented at each router hop.
Source Address	128 bits	IPv6 address of the sender.
Destination Address	128 bits	IPv6 address of the recipient.

Advantages and Disadvantages of IPv6

- **Advantages :**

- Support for security using (IPsec) Internet Protocol Security
- More powerful internet (128bit)
- Address allocation is done by the device itself

- **Disadvantages :**

- It will be much harder to remember IP addresses
- IPv6 is not available to machines that run IPv4
- Time to convert over to IPv6

IPv4 vs. IPv6

IPv4



12.143.7.212

- Numeric system
- 32-bit IP addresses
- Uses periods (.) to separate numbers
- Checksum fields
- Limited storage
- No built-in security
- Less suited to mobile devices
- Supported by millions of different devices
- Responsible for most internet communication

IPv6



2013:9ae7:0000:0000:
0000:rr02:0022:4434

- Alphanumeric system
- 128-bit IP addresses
- Uses colons (:) to separate numbers
- No checksum fields
- Unlimited storage
- Built-in Internet Protocol security
- Best suited to mobile devices
- Optionally supported by devices
- Increasing responsibility in internet communication

Advantages of IP

1. Universally Adopted
 - Used across almost all types of networks worldwide.
 - Standard protocol for communication over the internet.
2. Scalable
 - Supports hierarchical addressing.
 - Suitable for small, medium, and large networks.
3. Supports Internetworking
 - Enables communication between heterogeneous (different types of) networks.
 - Connects LANs, WANs, and other network types seamlessly.
4. Flexible
 - Allows coexistence of IPv4 and IPv6.
 - Supports gradual transition and upgrades.

Disadvantages of IP

1. No Reliability Guarantee
 - Does not guarantee delivery of packets.

- Does not ensure packets arrive in order.
 - Works on a "best-effort" delivery basis.
2. Address Exhaustion (IPv4)
 - Limited address space in IPv4.
 - Leads to shortage of available IP addresses.
 3. Fragmentation Overhead
 - Packet fragmentation in IPv4 can reduce performance.
 - Increases processing time and network overhead.
 4. Depends on Higher-Layer Protocols
 - Requires TCP or UDP for reliability and application support.
 - IP alone does not handle error recovery or flow control.

Subnetting

Subnetting is the process of dividing a large IP network into multiple smaller subnetworks (subnets) to improve performance, security, and efficient IP address utilization.

Instead of one large network, it creates logical segments that can be managed separately.

Purpose / Advantages of Subnetting

① Efficient Use of IP Address Space

- Prevents wastage of IP addresses.

② Improved Network Performance

- Smaller broadcast domains.
- Reduces unnecessary traffic.

③ Simplified Network Management

- Easier to monitor and troubleshoot.

④ Enhanced Security

- Segments departments/functions.
- Prevents unauthorized access between subnets.

Disadvantages of Subnetting

- Requires complex planning.
- Needs proper documentation.

- Additional router configuration required.

Subnetting Basics

An IPv4 address consists of:

- Network ID → Identifies the network portion
- Host ID → Identifies devices within the network

By borrowing bits from the Host ID, we create a Subnet ID.

Subnetting Formula Recap

1 Number of Subnets

$$\text{Number of Subnets} = 2^r$$

- r = number of bits borrowed from host portion

2 Hosts per Subnet

$$\text{Hosts per Subnet} = 2^h - 2$$

- h = remaining host bits
- Subtract 2 for:
 - Network address
 - Broadcast address

◆ IP Address Example (Class C)

IP Address: 192.168.1.0

Default Class: Class C

Default Subnet Mask: 255.255.255.0 (/24)

Supports:

$$2^8 - 2 = 254 \text{ hosts}$$

Step-by-Step Subnetting Example

Objective: Subnet **192.168.1.0/24** into **4 subnets**

Step 1: Calculate Subnet Bits

Required subnets = 4

$$2^2 = 4$$

👉 Borrow **2 bits** from host portion

Step 2: New Subnet Mask

Original Mask = /24

New Mask =

$$24 + 2 = /26$$

Dotted Decimal:

255.255.255.192

Step 3: Calculate Hosts per Subnet

Host bits remaining:

$$8 - 2 = 6$$

$$2^6 - 2 = 62 \text{ usable hosts per subnet}$$

Step 4: Subnet Ranges

Subnet No	Network Address	Usable IP Range	Broadcast Address
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Mask and Masking

♦ Subnet Mask

A **Subnet Mask** is a 32-bit number used to divide an IP address into: **Network Portion & Host Portion**

♦ Subnet Mask Format

- Written in **dotted decimal notation**
- Example: 255.255.255.0

♦ Purpose of Subnet Mask

- Distinguishes network part from host part.
- Performs a **bitwise AND operation** with the IP address.
- Extracts the **Network ID**.

♦ Masking (How It Works)

Masking = Applying subnet mask to IP address using **bitwise AND operation**.

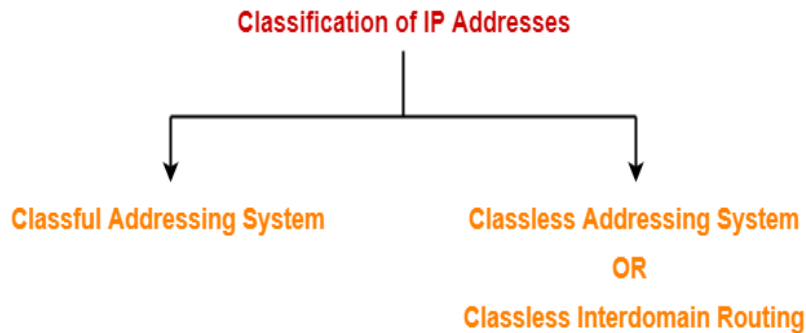
This helps determine:

- Which part is network
- Which part is host

Why Masking is Important

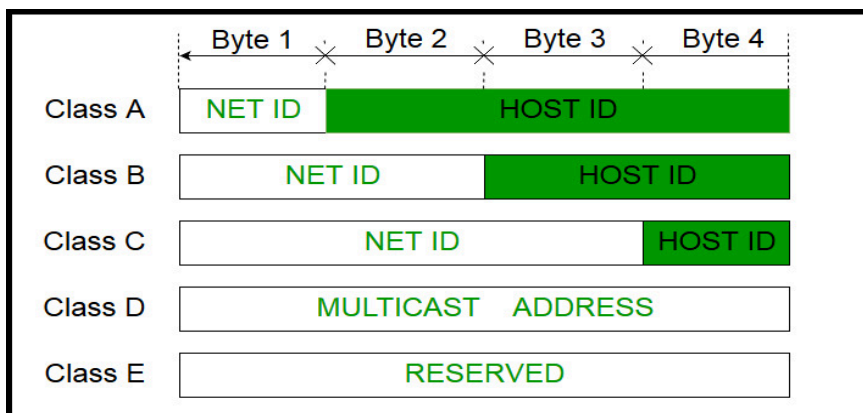
- Helps in subnetting (network segmentation).
- Prevents IP address wastage.
- Helps routers forward packets correctly.
- Enables hierarchical addressing.
- Simplifies network management.

IP Addressing System



Classful Addressing

- **Old system** used in the early days of IPv4.
- It divides IP addresses into fixed classes (A, B, C, D, E), each with predefined ranges.
- The class of an IP address determines the network portion and the host portion based on its class-specific subnet mask.



Class A

IP addresses belonging to class A are assigned to the networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.

The higher-order bit of the first octet in class A is always set to 0. The remaining 7 bits in the first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default subnet mask for Class A is 255.x.x.x. Therefore, class A has a total of:

- $2^{24} - 2 = 16,777,214$ host ID

IP addresses belonging to class A ranges from 0.0.0.0 - 127.255.255.255.



Class A

Class B

IP address belonging to class B is assigned to networks that range from medium-sized to large-sized networks.

- The network ID is 16 bits long.
- The host ID is 16 bits long.

The higher-order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine the network ID. The 16 bits of host ID are used to determine the host in any network. The default subnet mask for class B is 255.255.x.x. Class B has a total of:

- $2^{14} = 16384$ network address
- $2^{16} - 2 = 65534$ host address

IP addresses belonging to class B ranges from 128.0.0.0 – 191.255.255.255.



Class B

Class C

IP addresses belonging to class C are assigned to small-sized networks.

- The network ID is 24 bits long.
- The host ID is 8 bits long.

The higher-order bits of the first octet of IP addresses of class C is always set to 110. The remaining 21 bits are used to determine the network ID. The 8 bits of host ID are used to determine the host in any network. The default [subnet mask](#) for class C is 255.255.255.x. Class C has a total of:

- $2^{21} = 2097152$ network address
- $2^8 - 2 = 254$ host address

IP addresses belonging to class C range from 192.0.0.0 – 223.255.255.255.

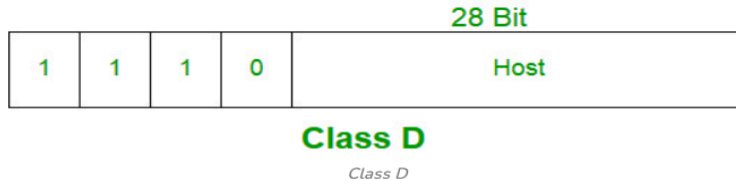


Class C

Class D

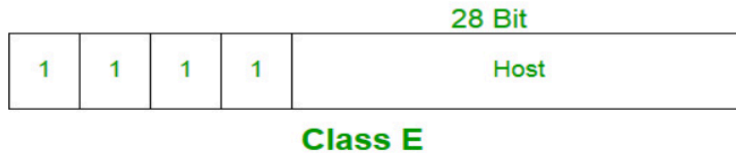
IP address belonging to class D is reserved for multi-casting. The higher-order bits of the first octet of IP addresses belonging to class D is always set to 1110. The remaining bits are for the address that interested hosts recognize.

Class D does not possess any subnet mask. IP addresses belonging to class D range from 224.0.0.0 – 239.255.255.255.



Class E

IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E range from 240.0.0.0 – 255.255.255.255. This class doesn't have any subnet mask. The higher-order bits of the first octet of class E are always set to 1111.



Structure of Classful Addressing

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	END ADDRESS
CLASS A	0	8	24	2^7 (128)	2^{24} (16,777,216)	0.0.0.0	127.255.255.255
CLASS B	10	16	16	2^{14} (16,384)	2^{16} (65,536)	128.0.0.0	191.255.255.255
CLASS C	110	24	8	2^{21} (2,097,152)	2^8 (256)	192.0.0.0	223.255.255.255
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.255
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.255

Disadvantages of Classful Addressing

Classes are fixed in size:

- Class A has ~16 million host addresses, even if you only need 500.
- Class C has only 254 usable hosts — too small for large networks.
- Result: Lots of unused addresses get locked inside networks.

Inflexible network sizes:

- You can't choose exactly how many bits to use for network vs host — you're stuck with A, B, or C sizes.
- No way to **customize subnet sizes** for different departments or offices.

Special Addresses in IPv4

Besides normal classes (A, B, C, D, E), IPv4 defines special-purpose addresses reserved by IANA for configuration, testing, and broadcasting.

❶ 0.0.0.0 – Default / Unspecified Address

Meaning: Represents "this network" or an unspecified address.

Usage:

- Used when a host has not yet been assigned an IP address.
- During boot time, a computer may send packets using 0.0.0.0 as the source address.

Purpose:

- Acts as a placeholder until a valid IP is assigned (e.g., during DHCP process).

❷ 127.0.0.1 – Loopback Address

Range: 127.0.0.0 – 127.255.255.255 (reserved for loopback)

Purpose:

- Allows a computer to send packets to itself.
- Used for testing TCP/IP stack.

Example Command: ping 127.0.0.1

Key Point:

- Packets sent to this address **never leave the host machine**.
- No physical network is used.

❸ 255.255.255.255 – Limited Broadcast Address

Meaning: Sends message to all devices in the local network segment.

Usage:

- Used when device wants to communicate with all devices on local network.
- Example: ARP requests.

Important Note:

- Not forwarded by routers.
- Limited to the local network only.

4 Private IPv4 Address Ranges

Private IPs:

- Used inside local networks.
- Not routable on the Internet.
- Help conserve public IP addresses.

Private Address Ranges

Range	Class	Example
10.0.0.0 – 10.255.255.255	Class A	10.1.1.1
172.16.0.0 – 172.31.255.255	Class B	172.16.5.4
192.168.0.0 – 192.168.255.255	Class C	192.168.1.1

Classless Addressing

- **Modern system** (introduced in 1993).
- Classless Inter-Domain Routing (CIDR) is a method for efficiently allocating IP addresses and routing Internet Protocol (IP) packets.
- Unlike classful addressing, which divides IP addresses into fixed classes (A, B, C, etc.), CIDR allows for variable-length subnet masks (VLSM).
- This means that networks can be divided into smaller, more flexible subnets according to their specific needs, rather than being constrained by predefined class boundaries.

CIDR Notation

In CIDR subnet masks are denoted by /X. For example a subnet of 255.255.255.0 would be denoted by /24. To work a subnet mask in CIDR, we have to first convert each octet into its respective binary value. For example, if the subnet is of 255.255.255.0. then :

First Octet

255 has 8 binary 1's when converted to binary

Second Octet

255 has 8 binary 1's when converted to binary

Third Octet

255 has 8 binary 1's when converted to binary

Fourth Octet

0 has 0 binary 1's when converted to binary

Therefore, in total there are 24 binary 1's, so the subnet mask is /24.

While creating a network in CIDR, a person has to make sure that the masks are contiguous, i.e. a subnet mask like 10111111.X.X.X can't exist.

- All the 1 bits (network bits) come first.
- All the 0 bits (host bits) come after.
- You cannot mix them (no 1s after 0s in the mask).

With CIDR, we can create Variable Length Subnet Masks, leading to less wastage of IP addresses. It is not necessary that the divider between the network and the host portions is at an octet boundary. For example, in CIDR a subnet mask like 255.224.0.0 or 11111111.11100000.00000000.00000000 can exist.

Network Address Translation (NAT)

Network Address Translation (NAT) is a technique used in computer networks to map private IP addresses to public IP addresses and vice versa. It is widely used in IPv4 networks to overcome the problem of limited public IP addresses.

Why NAT is Needed

1. IPv4 Address Shortage

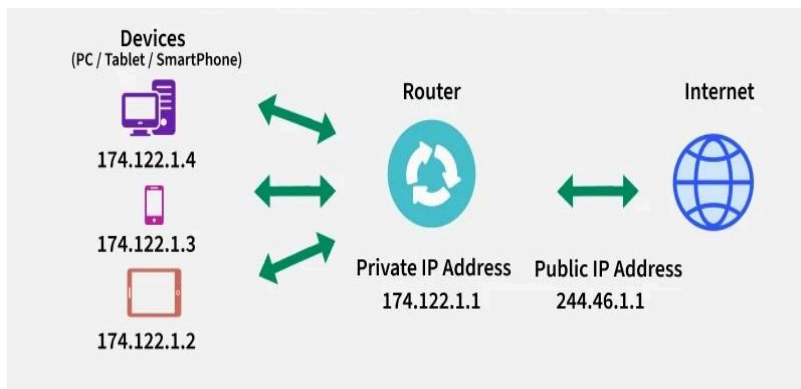
- Only about **4.3 billion IPv4 addresses** exist.
- With billions of devices, this space is insufficient.
- NAT allows many private devices to share a **single public IP**.

2. Security and Privacy

- Internal devices are hidden from the Internet.
- Only the router's public IP is visible externally.

3. Flexibility

- Organizations can change internal IP addresses without affecting external communication.



In network address translation, a network device, often a router or NAT firewall, assigns a computer or computers inside a private network a public address. In this way, network address translation allows the single device to act as an intermediary or agent between the local, private network and the public network that is the internet. NAT's main purpose is to conserve the number of public IP addresses in use, for both security and economic goals.

How NAT Works

1. Device initiates a connection (e.g., to a website).
2. NAT router replaces the source IP (private) with the public IP, and assigns a unique source port. NAT stores this mapping in a NAT table.
3. When the response comes back, the router uses this mapping to forward the data to the correct internal device.

Example: NAT in Action

Company LAN private IPs:

- 192.168.1.10
- 192.168.1.20
- 192.168.1.30

Public IP of router:

- 203.0.113.25

Translation:

- 192.168.1.10:1025 → 203.0.113.25:5001
- 192.168.1.20:1030 → 203.0.113.25:5002
- 192.168.1.30:1035 → 203.0.113.25:5003

To the outside world, all traffic appears from **203.0.113.25**.

Types of NAT

1. Static NAT (One-to-One Mapping)

- Each private IP is permanently mapped to one public IP.
- Used for hosting servers.
- Example: 192.168.1.10 → 203.0.113.50

2. Dynamic NAT (Many-to-Many Mapping)

- Uses a pool of public IP addresses.
- Private IPs are mapped to any available public IP dynamically.
- Mapping changes over time.

3. Port Address Translation (PAT) / NAT Overload

- Most common type (used in home routers).
- Many private IPs share a **single public IP**.
- Differentiation is done using **port numbers**.

Example:

- 192.168.1.2:1030 → 203.0.113.25:5001
- 192.168.1.3:1045 → 203.0.113.25:5002

NAT in IPv6

- IPv6 provides a huge address space (**2^{128} addresses**).
- NAT is generally **not required** in IPv6.
- IPv6 uses:
 - **Global Unicast Addresses** (Internet communication)
 - **Unique Local Addresses (ULAs)** (private networks)

However, **NAT66** or **Prefix Translation** may be used in special cases (renumbering, privacy, provider independence).

Advantages of NAT

- Conserves IPv4 public address space.
- Provides internal network security.
- Flexibility in IP addressing.
- Supports multiple devices with one public IP (via PAT).

Disadvantages of NAT

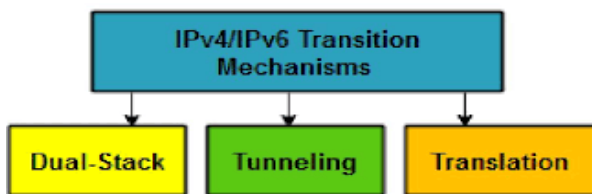
- Introduces processing overhead.
- Breaks end-to-end connectivity.
- Makes peer-to-peer communication difficult.
- Some protocols (VoIP, FTP, IPsec) may face issues.

Transition from IPv4 to IPv6

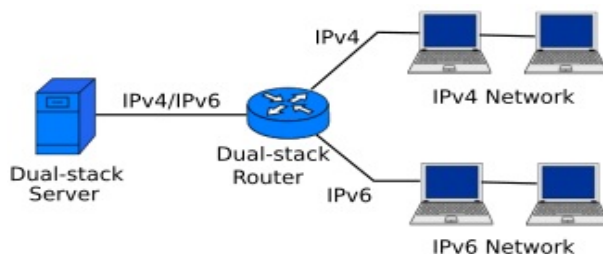
Why IPv6?

- IPv4 uses 32-bit addresses → only about 4.3 billion IPs.
- Due to the growth of internet users and devices, IPv4 addresses are running out.
- IPv6 uses 128-bit addresses → allows trillions of unique IPs.

Various organizations are currently working with IPv4 technology and in a very short time, we can not switch directly from IPv4 to IPv6. Instead of only using IPv6, we use a combination of both and transition means not replacing IPv4 but co-existing of both.



Dual Stack



Dual Stack means that a device (like a router, server, or end-user computer) is configured with both an IPv4 and an IPv6 address. It can communicate using either protocol, depending on what the destination supports.

How It Works – Step by Step:

1. Dual IP Addresses

- The device has two IP addresses:
 - One IPv4 address (e.g., 192.168.1.2)
 - One IPv6 address (e.g., 2001:db8::1)

2. DNS Resolution

- When the device wants to connect to a website or service:
 - It sends a DNS query.
 - If the domain has both A (IPv4) and AAAA (IPv6) records, it receives both.

3. Protocol Selection (Happy Eyeballs Algorithm)

- The system prefers IPv6 if available and functional.
- If IPv6 doesn't work or is slow, it falls back to IPv4.
 - This is managed by the Happy Eyeballs algorithm (trying both IPv6 and IPv4 at the same time or almost the same time — and choosing the one that responds first).

4. Communication

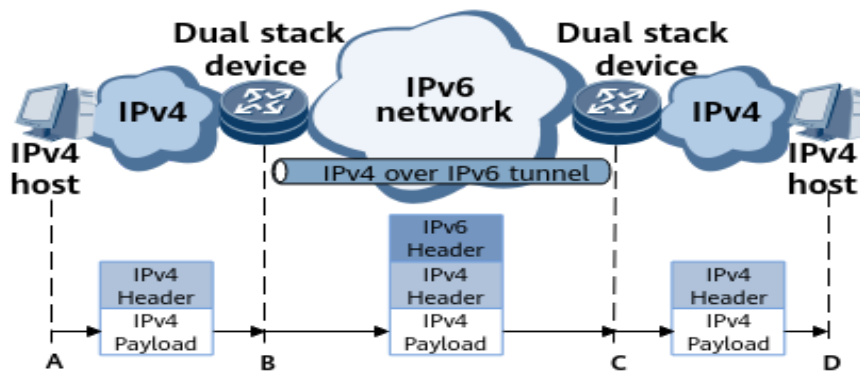
- The device uses IPv6 to communicate with IPv6-enabled services.
- It uses IPv4 when communicating with IPv4-only devices or services.

Example Scenario:

You visit www.example.com:

- DNS returns:
 - 192.0.2.1 (IPv4)
 - 2001:db8::1 (IPv6)
- Your device **tries IPv6 first**.
- If IPv6 connects, it's used.
- If not, it **falls back to IPv4**.

Tunneling



Tunneling is a technique used to enable communication between IPv4 networks over an IPv6-only backbone or vice versa.

How Tunneling Works:

1. Encapsulation:

When an IPv4 packet needs to travel across an IPv6 network, it's encapsulated within an IPv6 packet. This means the IPv4 packet becomes the payload of the IPv6 packet.

2. Transmission:

The encapsulated IPv6 packet is then routed through the IPv6 network.

3. Decapsulation:

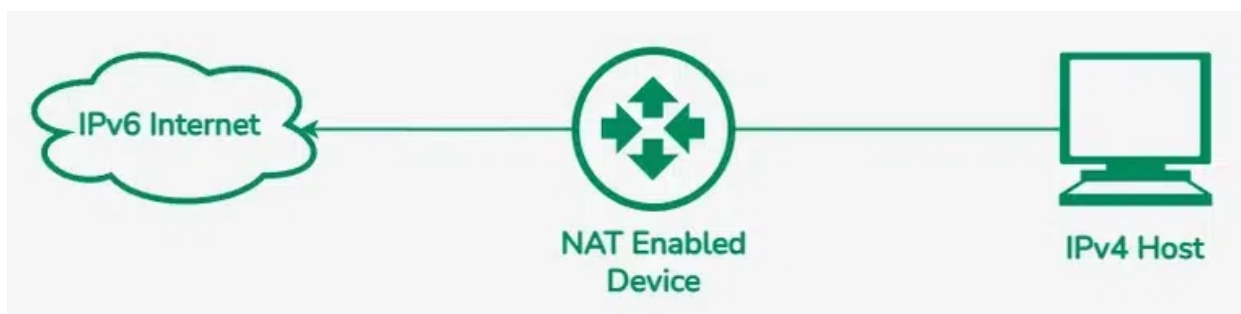
At the destination, the IPv6 packet is received, and the IPv4 packet is extracted (decapsulated).

4. Forwarding:

Finally, the IPv4 packet is forwarded to its final destination within the IPv4 network through IPv6 network.

Types of Tunneling :- Manual (preconfigured) & Automatic (6 to 4)

Translation



Network Address Translation Protocol Translation (NAT-PT), is a technique used to enable communication between IPv4 and IPv6 networks by translating one protocol to the other.

A host with IPv4 address sends a request to an IPv6 enabled server on the Internet that does not understand IPv4 address. In this scenario, the NAT-PT device can help them communicate. When the IPv4 host sends a request packet to the IPv6 server, the NAT-PT device/router strips down the IPv4 packet, removes IPv4 header, and adds IPv6 header and passes it through the Internet. When a response from the IPv6 server comes for the IPv4 host, the router does vice versa.

Techniques:- NAT64 & DNS64

IPv4 ↔ IPv6 (in protocol translation)

- Example: NAT64 maps an IPv4 address to an IPv6 address so an IPv4-only host can talk to an IPv6-only host.
- This can be done:
 - Stateless → use a fixed formula (embed IPv4 bits into an IPv6 address).
 - Stateful → store mappings in a table for active sessions.

Address Mapping

In computer networks, communication requires two types of addresses:

❶ Logical Address (IP Address)

- Used by the **Network Layer**
- Identifies devices across networks
- Example: 192.168.1.5

❷ Physical Address (MAC Address)

- Used by the **Data Link Layer**
- Used for frame delivery within a LAN
- 48-bit address (e.g., 08:00:27:65:43:1F)

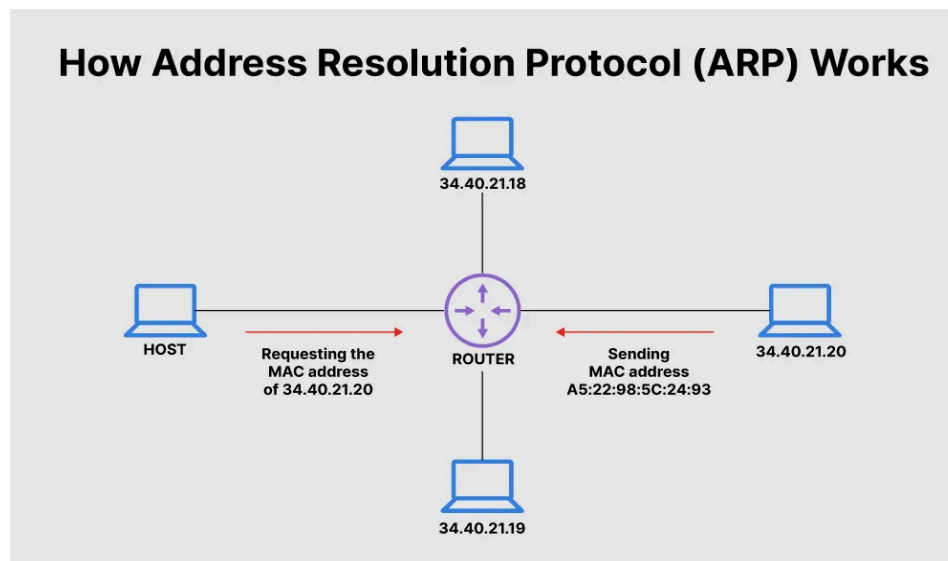
Mapping logical address to physical address is called **Address Mapping**.

Different protocols handle mapping:

- IPv4 → **ARP**
- IPv6 → **NDP**

ARP (Address Resolution Protocol)

- Used in **IPv4 networks**
- Maps **32-bit IPv4 address** → **48-bit MAC address**
- Required for communication inside a LAN



1. Host checks **ARP cache** (IP–MAC table).
2. If not found, it sends an **ARP Request (broadcast)**:
"Who has IP X.X.X.X?"
3. The device with that IP replies with its **MAC address (ARP Reply)**.
4. Sender stores mapping in ARP cache.

Types of ARP

1. **Proxy ARP**
 - Router answers ARP on behalf of another device.
2. **Gratuitous ARP**
 - Host announces its own IP–MAC mapping.
3. **Reverse ARP (RARP)**

- Maps MAC address → IP address.

◆ Advantages of ARP

- Smooth communication between IP & Ethernet/Wi-Fi.
- No manual MAC configuration needed.

◆ Disadvantages of ARP

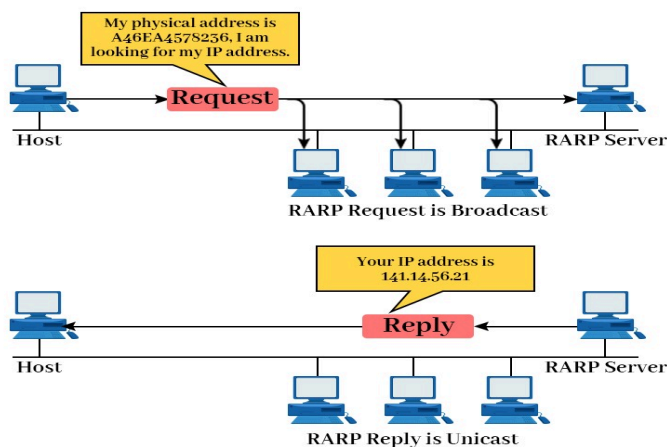
- Vulnerable to **ARP spoofing/poisoning**.
- Works only within a single broadcast domain.

RARP (Reverse Address Resolution Protocol)

- Opposite of ARP.
- Maps **MAC address** → **IP address**.
- Mainly used by **diskless workstations**.

How RARP Works

1. Device sends **RARP Request (broadcast)** with its MAC.
2. **RARP Server** replies with assigned IP address.
3. Device uses that IP to join the network.



◆ Advantages of RARP

- Useful for diskless machines.
- Automatic IP assignment.

◆ **Disadvantages of RARP**

- Requires dedicated RARP server on each LAN.
- Supports only IPv4.
- Replaced by **BOOTP** and **DHCP** (more flexible).

NDP (Neighbor Discovery Protocol – IPv6)

- IPv6 replacement for:
 - ARP
 - RARP
 - Some ICMP functions
- Works at **Network Layer**
- Uses **ICMPv6 messages**

◆ **Functions of NDP**

1. **Address Resolution**
Maps IPv6 → MAC (replaces ARP)
2. **Neighbor Unreachability Detection (NUD)**
Checks if neighbor is reachable.
3. **Router Discovery**
Hosts automatically discover routers.
4. **Prefix Discovery**
Hosts learn subnet prefixes.
5. **Duplicate Address Detection (DAD)**
Ensures no two devices use same IPv6 address.

Internetworking

Internetworking is the process of connecting multiple computer networks to function as one large communication system.

- Best example: Internet
- Connects LANs, WANs, Wi-Fi, cellular, satellite networks.
- Uses standardized protocols (TCP/IP).

Role of Network Layer

The Network Layer is responsible for:

- Logical addressing
- Packet forwarding
- Fragmentation
- Error handling
- Diagnostics
- Routing

Logical Addressing

Every device in an internetwork must have a unique logical address.

- Implemented using IP addresses (IPv4 & IPv6).
- Independent of physical (MAC) address.

Functions

- Identifies source and destination across networks.
- Hardware independent.
- Supports hierarchical addressing for routing.

Example

Device A (192.168.1.10) in LAN1 communicates with
Device B (172.16.5.20) in LAN2 using IP addressing.

Packet Forwarding

Packet forwarding is the actual movement of packets from router input port to correct output port.

Steps in Packet Forwarding

1. Router receives packet.
2. Extracts destination IP.

3. Searches Forwarding Information Base (FIB).
4. Forwards packet via appropriate interface.

Techniques Used

1. Longest Prefix Match (LPM)

- Chooses the most specific route (CIDR-based).

Example:

Routing table contains:

- 192.168.10.0/24 → Interface 2
- 192.168.0.0/16 → Interface 3

Destination: 192.168.10.25

Router selects /24 (longest prefix) → Interface 2.

2. Equal-Cost Multi-Path (ECMP)

- Uses multiple paths with equal cost.
- Provides load balancing.

Fragmentation and Reassembly

Different networks support different **MTU (Maximum Transmission Unit)** sizes.

If packet size > MTU → packet must be fragmented.

Fragmentation

- Occurs at sender or intermediate router.
- Packet split into fragments.
- Each fragment has its own header.

Fragment Header Fields:

- Identification – Links fragments.
- Fragment Offset – Position of fragment.
- MF (More Fragments) flag –

- 1 = More fragments follow
- 0 = Last fragment

Reassembly

- Performed only at destination.
- Uses Identification + Offset fields.
- Rebuilds original packet.

Example

- MTU = 1500 bytes
- Packet size = 4000 bytes

Fragments $\approx$

- 1480 bytes
- 1480 bytes
- 1040 bytes

(Data size only, excluding header)

Advantages

- Enables communication across networks with different MTUs.

Disadvantages

- Increased overhead (more headers).
- If one fragment is lost, entire packet must be retransmitted.

Error Handling & Diagnostics

IP is a **best-effort protocol** → does NOT guarantee error-free delivery.

Errors handled using additional mechanisms.

Error Handling Mechanisms

1. Checksums

- **IPv4:** Header includes checksum.
- **IPv6:** No header checksum (handled by upper layers).

2. ICMP (Internet Control Message Protocol)

Used for:

- Sending error messages
- Network diagnostics

Examples:

- Destination Unreachable
- Time Exceeded (used in traceroute)
- Echo Request/Reply (used in ping)

Diagnostic Tools

1. Ping

- Uses ICMP Echo Request/Reply.
- Tests reachability.

2. Traceroute

- Determines path taken by packets.
- Uses TTL & ICMP Time Exceeded messages.

3. Packet Analyzers (e.g., Wireshark)

- Capture and analyze packets in detail.

Advantages of Diagnostic Tools

- Maintain network reliability.
- Help troubleshoot large networks.

Disadvantages of Diagnostic Tools

- Can be exploited (e.g., ping flood attack).
- May reveal network structure to attackers.

ICMP (Internet Control Message Protocol)

- Network layer protocol used in **IPv4** to:
 - Report errors
 - Send control messages

Function

- ICMP **does NOT fix errors**
- It only **notifies the source** so higher layers (like TCP) can take action.

Encapsulation

- ICMP messages are **carried inside IP packets**.

Uses of ICMP

1. Error reporting
 - Destination unreachable
 - Time exceeded
 - Parameter problem
2. Network diagnostics
 - Used by **ping** and **traceroute**
3. Control messages
 - Redirect, etc.

ICMPv6

Successor of ICMP for **IPv6 networks**

- Performs ICMP functions + extra features

Additional Features

- **Neighbor Discovery Protocol (NDP)**
→ Replaces ARP and RARP
- **Multicast Listener Discovery (MLD)**
→ Manages multicast group membership
- Optimized error reporting for IPv6

Improvements over ICMPv4

- Supports IPv6 large address space
- Uses multicast-based communication
- Provides **Secure Neighbor Discovery (SEND)** for spoofing protection
- Handles IPv6-specific errors like **Packet Too Big**

Common ICMP Error Messages

❶ Destination Unreachable

Occurs when packet cannot reach destination.

Types:

- **Network unreachable** → No route to network
- **Host unreachable** → Destination device unavailable
- **Port unreachable** → Application port closed
- **Protocol unreachable** → Protocol not supported

Example: Connecting to a closed TCP port → ICMP sends "Port Unreachable".

❷ Time Exceeded

- Sent when **TTL (IPv4)** or **Hop Limit (IPv6)** becomes 0
- Prevents infinite routing loops
- Used by **traceroute** to map path

❸ Parameter Problem

- Sent when IP header has errors or invalid field

- Packet cannot be processed
- Example: Unknown IPv6 extension header

Multicasting

Communication method where **one sender** → **multiple selected receivers**

- More efficient than sending separate copies

Characteristics

- Data sent once, delivered to all group members
- Used in:
 - Video conferencing
 - Live streaming
 - Online gaming

Multicast Protocol

IGMP (Internet Group Management Protocol)

- Used in **IPv4 networks**
- Manages host membership in multicast groups
- Hosts inform routers when they want to:
 - Join a multicast group
 - Leave a multicast group

Working

- 1. Joining a Multicast Group:** When a host wants to receive multicast data (e.g., live video stream), it sends a Join message to the local router.
- 2. Router Tracks Active Members:** The router periodically sends IGMP Membership Query messages to all hosts in the network. These queries ask:
"Is anyone still interested in this multicast group?"
- 3. Leaving a Multicast Group:** When a host no longer wants to receive multicast data, it sends an IGMP Leave message to the router.

Example

In an IPv4 IPTV service:

- Users join a TV channel → IGMP Join
- Router checks active viewers → Query
- User changes channel → Leave
- Only subscribed users receive the stream

MLD (Multicast Listener Discovery)

- Used in IPv6 networks.
- Works as the IPv6 version of IGMP.
- It operates as part of **ICMPv6**, not as a separate protocol.

Working

- 1. Joining a Multicast Group:** When a host wants to receive multicast traffic, it sends an MLD Report message.
- 2. Router Checks Active Groups:** The router periodically sends MLD Query messages. These queries verify which multicast groups are still active.
- 3. Leaving a Multicast Group:** When a host leaves a multicast group, it sends an MLD Done message.

Example:

Scenario: IPv6 Video Conference in a Company Network

- Employees join a live video meeting on the company network.
- Each employee's computer sends an **MLD Report message** to the local IPv6 router saying:
"I want to receive this multicast video stream."
- The router records which devices want the stream.
- The video server sends **one multicast stream**, and the router forwards it **only to those devices** that joined.
- When an employee leaves the meeting, their device sends an **MLD Done message**.

Applications of Multicasting

- 1. IPTV (Internet Protocol Television):** Delivers live TV channels to multiple users without sending separate unicast streams.
- 2. Video Conferencing & Online Meetings:** Platforms like Zoom or Teams use multicast for real-time audio and video streaming.
- 3. Online Gaming:** Multiplayer games distribute real-time updates (player moves, scores) to all participants.
- 4. Software Updates:** Organizations multicast large updates (e.g., OS patches) to multiple systems at once.
- 5. Stock Market Feeds:** Financial institutions distribute real-time stock quotes to multiple clients.
- 6. Distance Learning:** Online classrooms stream lectures to multiple students simultaneously.

Data Delivery

The **Network Layer** is responsible for delivering packets from the source device to the correct destination device.

Types of Delivery

1. Unicast Delivery

Unicast is a **one-to-one communication** method where a single packet is sent from one source to exactly one destination host.

Features:

- Most common type of data delivery on the Internet.
- Requires a unique destination IP address.
- Efficient for point-to-point communication.

Examples:

- Sending an email
- Browsing a website
- Downloading a file

2. Multicast Delivery

Multicast is a **one-to-many communication** method where a single packet is delivered to a group of hosts that have joined the multicast group.

Features:

- Uses special multicast IP address ranges:
 - IPv4: **224.0.0.0 – 239.255.255.255**
 - IPv6: **FF00::/8**
- Only receivers who join the multicast group receive the data.
- Saves bandwidth compared to sending multiple unicast packets.

Examples:

- Streaming a live lecture to multiple students
- Distributing stock market updates

3. Broadcast Delivery (IPv4 only)

Broadcast is a **one-to-all communication** method where a packet is delivered to all hosts within a local network (subnet).

Features:

- Available only in IPv4 (IPv6 does not support broadcast).
- Uses broadcast IP address (e.g., **255.255.255.255** for limited broadcast).
- Useful for service discovery and network announcements.
- Can cause network congestion if overused.

Example:

- A DHCP server broadcasting an IP address offer to all hosts in the local network.

4. Anycast Delivery (IPv6 only)

Anycast is a **one-to-nearest communication** method where a packet is delivered to the nearest host (based on routing distance) among a group of hosts sharing the same IP address.

Features:

- Available only in IPv6.
- Multiple devices (e.g., servers) can share the same anycast IP address.
- Routing protocol selects the closest destination automatically.
- Improves load balancing and reliability.

Example:

- A client accessing the nearest DNS root server using anycast addressing.

Routing

Routing is the process of determining the optimal path for data packets from source to destination across interconnected networks.

Goals

- Efficient delivery
- Reliability
- Minimum delay
- Reduced congestion
- Lower network cost

How Routing Works

1. Host sends packet.
2. Packet reaches router.
3. Router checks routing table.
4. Packet is forwarded via selected path.

Types of Routing

1. Static Routing

- Manually configured.
- Simple but not scalable.
- Used in small networks.

2. Dynamic Routing

- Routes learned automatically.
- Uses routing protocols.

Examples:

- RIP (Routing Information Protocol) – Uses hop count.
- OSPF (Open Shortest Path First) – Uses link-state information.
- BGP (Border Gateway Protocol) – Used between ISPs.

Advantages of Routing

- Efficient packet delivery.
- Supports large and complex networks.

Disadvantages of Routing

- Congestion may cause delays.
- Misconfiguration can create routing loops.

Techniques for Choosing Network Paths

- One of the primary functions of the Network Layer is routing.
- Routing decides the best path that a packet should take to reach its destination.
- Since multiple paths may exist, routing algorithms help in selecting the most efficient one.

Three Major Path Selection Techniques

1. Distance Vector Routing
2. Link State Routing
3. Dijkstra's Algorithm (Shortest Path First – SPF)

Distance Vector Routing

- Each router knows only its immediate neighbors.

- Distance usually measured in hop count (can also be delay, bandwidth, etc.).
- Each router maintains a routing table.

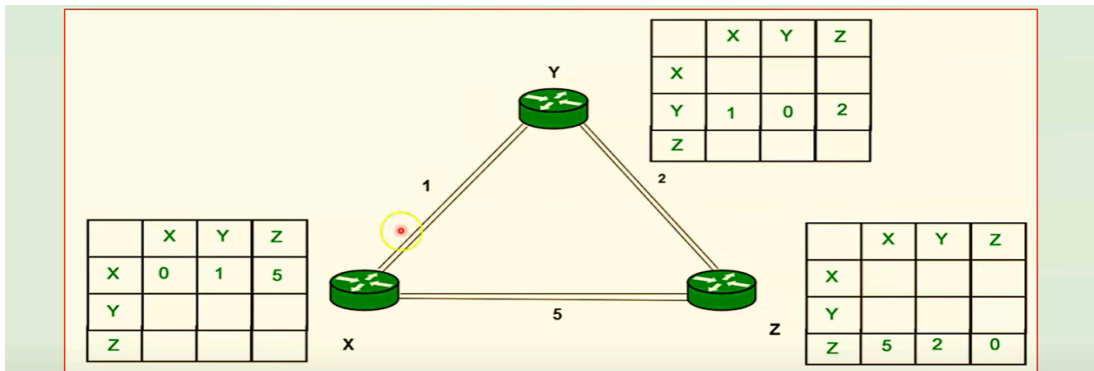
Routing Table Contains:

- Destination network
 - Distance (cost)
 - Next-hop router
- When a node x receives new DV estimate from any neighbor v, it saves v's distance vector and it updates its own DV using B-F equation:

$$D_x(y) = \min \{ C(x,v) + D_v(y), D_x(y) \} \text{ for each node } y \in N$$

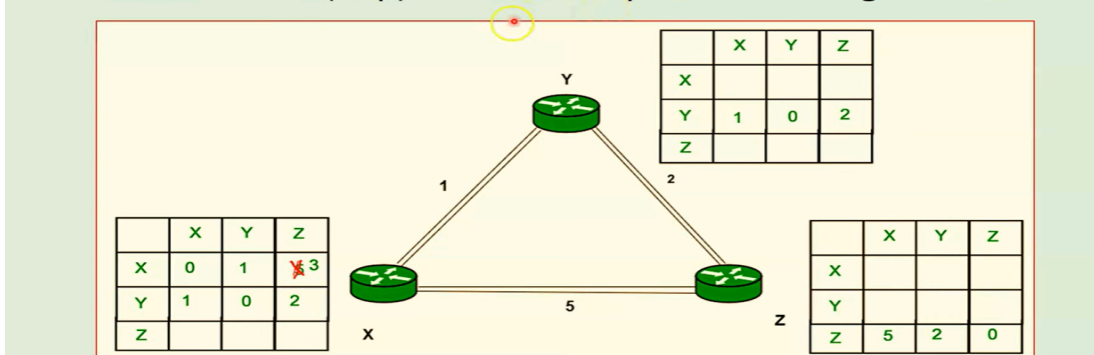
Example

- Consider 3-routers X, Y and Z as shown in figure. Each router have their routing table. Every routing table will contain distance to the destination nodes.

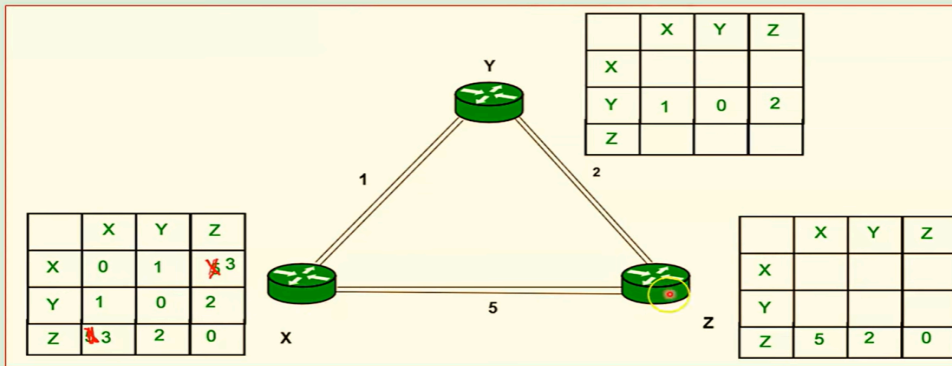


- Consider router X, X will share its routing table to neighbors and neighbors will share their routing table to X and distance from node X to destination will be calculated using Bellman-Ford equation.

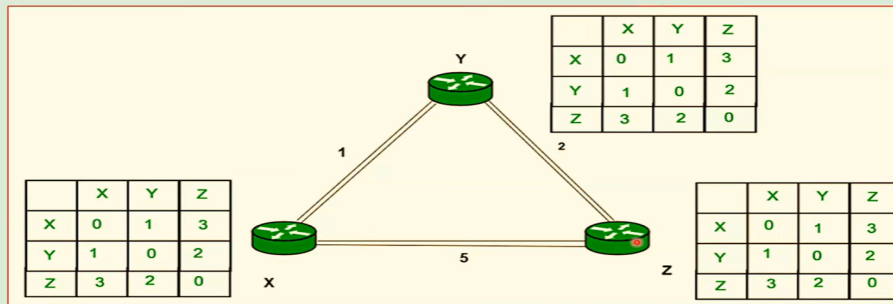
- As we can see that distance will be less going from X to Z when Y is an intermediate node (hop) so it will be updated in routing table X.



Similarly for Z also



Finally the routing table for all



- Distance Vector routing uses UDP(User datagram protocol) for transportation

Problems

- **Count-to-Infinity Problem:** Distance keeps increasing if route fails.
- **Slow Convergence:** Updates take time in large networks.

Protocols Using Distance Vector

- RIP (Routing Information Protocol)
- IGRP (Interior Gateway Routing Protocol)

Advantages

- Simple to implement
- Low memory & CPU usage

Disadvantages

- Slow convergence
- Routing loops possible
- Not scalable for large networks

Link State Routing

Each router:

- Knows all its neighbors
- Knows cost of each link
- Builds a complete **network topology map**
- Routers exchange information using: **LSAs (Link State Advertisements)**

How It Works

- 1. Neighbor Discovery** – Identify directly connected routers.
- 2. Link Cost Measurement** – Based on bandwidth, delay, etc.
- 3. Flooding LSAs** – Share info with all routers.
- 4. Topology Database** – All routers build identical network map.
- 5. Shortest Path Calculation** – Run Dijkstra's algorithm.

Protocols Using Link State

- OSPF (Open Shortest Path First)
- IS-IS (Intermediate System to Intermediate System)

Advantages

- Faster convergence than Distance Vector
- Loop-free routing
- Better scalability (large networks)

Disadvantages

- Complex configuration
- Higher CPU & memory usage

Dijkstra's Algorithm (Shortest Path First – SPF)

Used by Link State protocols like OSPF.

Network Representation

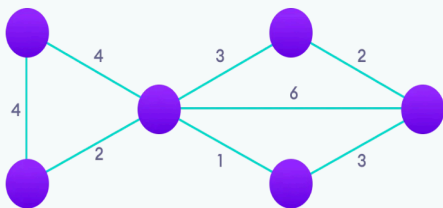
- Routers → Nodes
- Links → Edges
- Costs → Weights

Algorithm Steps

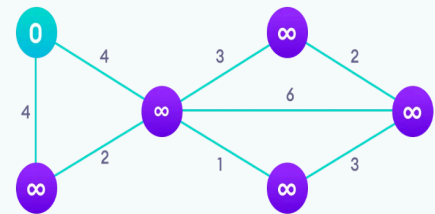
1. Assign distance to every node:
 - Source = 0
 - Others = ∞
2. Mark all nodes unvisited.
3. Select unvisited node with smallest distance.
4. For each neighbor:
 - Calculate new tentative distance.
 - Update if smaller.
5. Mark node as visited.
6. Repeat until all nodes visited.

Example of Dijkstra's algorithm

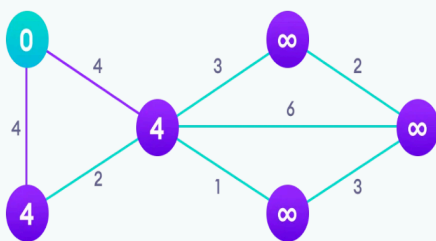
It is easier to start with an example and then think about the algorithm.



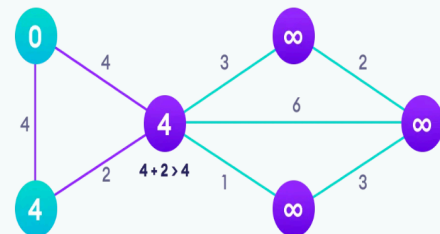
Start with a weighted graph



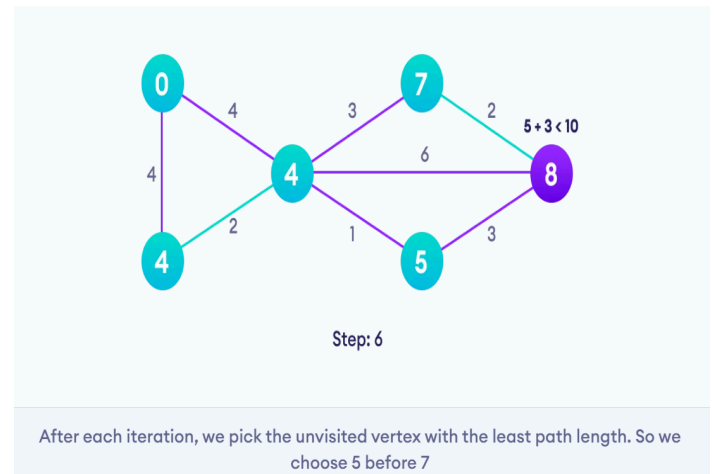
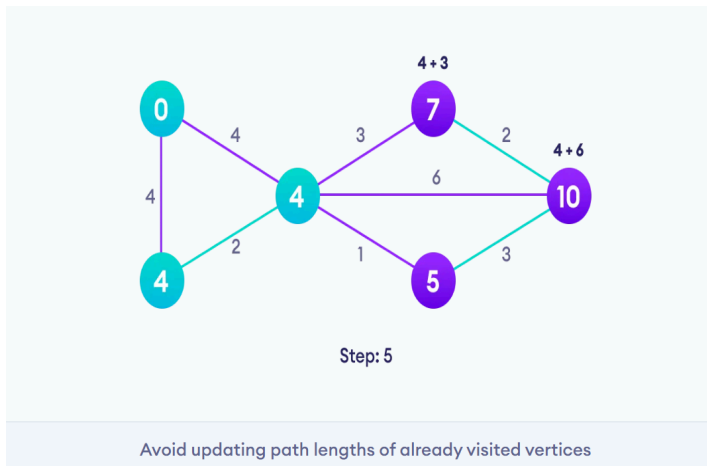
Choose a starting vertex and assign infinity path values to all other devices



Go to each vertex and update its path length



If the path length of the adjacent vertex is lesser than new path length, don't update it



Advantages

1. Guarantees shortest paths
2. Eliminates routing loops
3. Efficient for modern routing protocols

Disadvantages

1. Computationally expensive in very large networks
2. Requires full knowledge of network topology

Model Questions

3 Mark Questions

1. Define the Network Layer.
2. What is the role of logical addressing?
3. What is packet switching?
4. Differentiate between virtual circuit and datagram networks.
5. What is a repeater?
6. What is the function of a hub?
7. Mention any three functions of a router.
8. What is a gateway?
9. Define IPv4 address.
10. What is subnetting?

6 Mark Questions

1. Explain the key functions of the Network Layer.
2. Describe packet switching with an example.
3. Compare virtual circuit and datagram networks.
4. Describe the working of a bridge.
5. Explain how a router forwards packets.
6. Explain IPv4 address classes with examples.
7. Describe the purpose and working of subnet masks.
8. Explain classful vs. classless addressing.
9. Describe IPv6 features and benefits.
10. Explain NAT and its types.

10 Mark Questions

1. Explain different networking and internetworking devices and their functions.
2. Describe IPv4 addressing, special addresses, subnetting, and CIDR.
3. Explain IPv6 addressing, structure, features, and types of IPv6 addresses.
4. Explain IP packet structure, fragmentation, and reassembly in IPv4 and IPv6.
5. Describe IP-related protocols: ARP, RARP, ICMP, and IGMP.